Parallelization of Diophantine Solvers

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Introduction

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Answer: Yes! The smallest was found in 2002:

$$1117^6 + 770^6 = 1092^6 + 861^6 + 602^6 + 212^6 + 84^6$$

Resta-Meyrignac Algorithm

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Finally, we define $v = a_1^6 + a_2^6 - (b_1^6 + b_2^6)$ and check if we can decompose $v/7^6 = c_1^6 + c_2^6 + c_3^6$.

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OpenMP Implementation

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For comparison, the 2002 paper mentions it took about 5 minutes to obtain just the single smallest solution.

CUDA Implementation: Host Setup

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Before launching the kernel, the CPU builds two lookup tables:

- a residue table modulo $7^6 = 117649$ for possible b_2 values;
- a sorted table of pair sums $x^6 + y^6$ used later to recognize two of the remaining three terms.

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- a sorted table of pair sums $x^6 + y^6$ used later to recognize two of the remaining three terms.

These tables are copied once to GPU memory with `cudaMemcpy`, so all GPU threads can reuse the same precomputed data.

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This gives a simple load distribution without storing all (a_1, a_2) pairs explicitly.

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For each assigned (a_1, a_2), the kernel computes

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Additional modular filters modulo $13, 19, 27, 31, 32, 49$ remove values that cannot be written as a sum of three sixth powers.

CUDA Implementation: Final Decomposition

After choosing b_1 and b_2, the remaining value is

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If this succeeds, the stored solution is $(a_1, a_2, b_1, b_2, 7c_1, 7c_2, 7c_3)$.

CUDA Implementation: Collecting Results

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After the kernel finishes:

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- solutions are sorted before printing;
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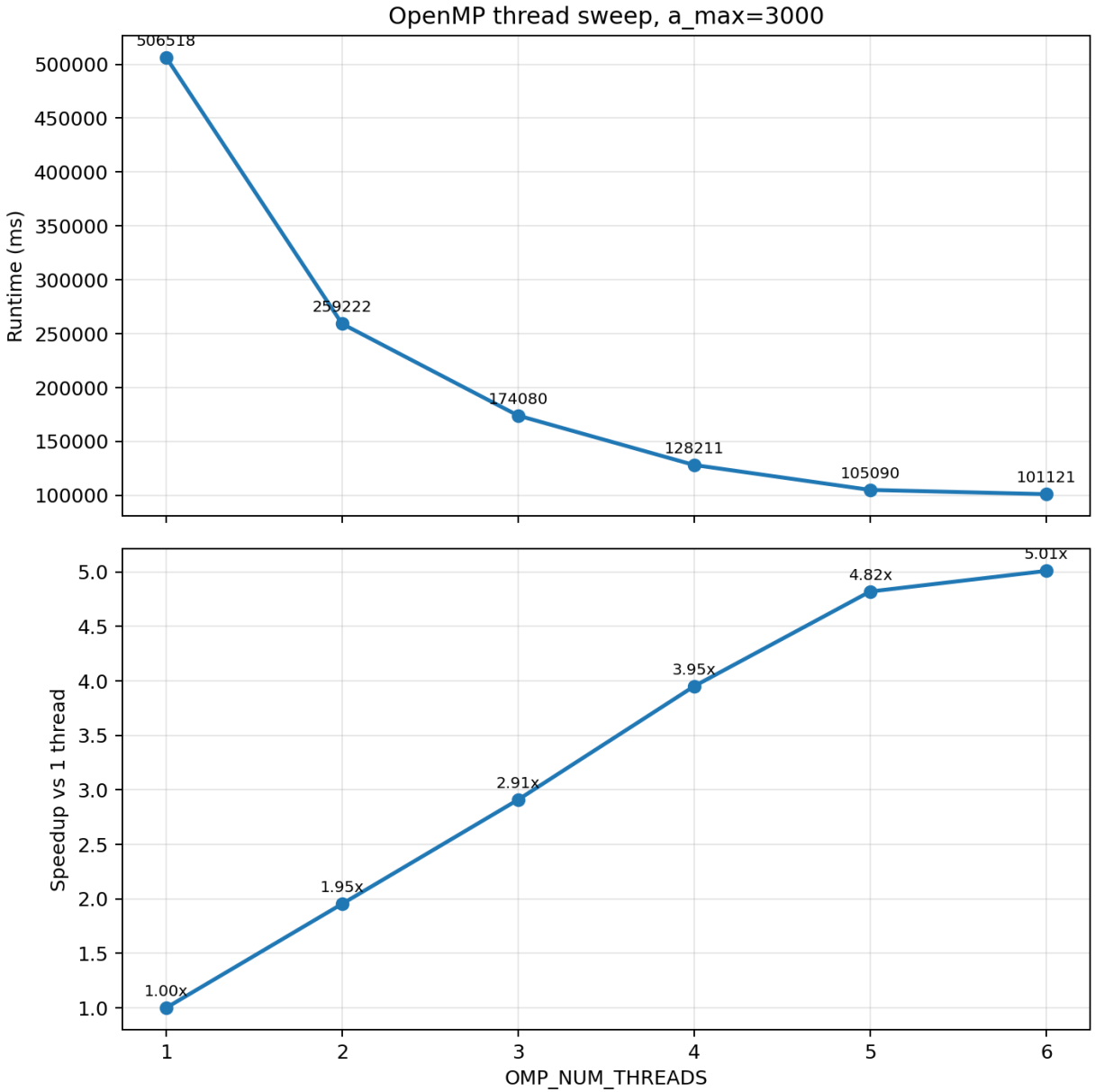
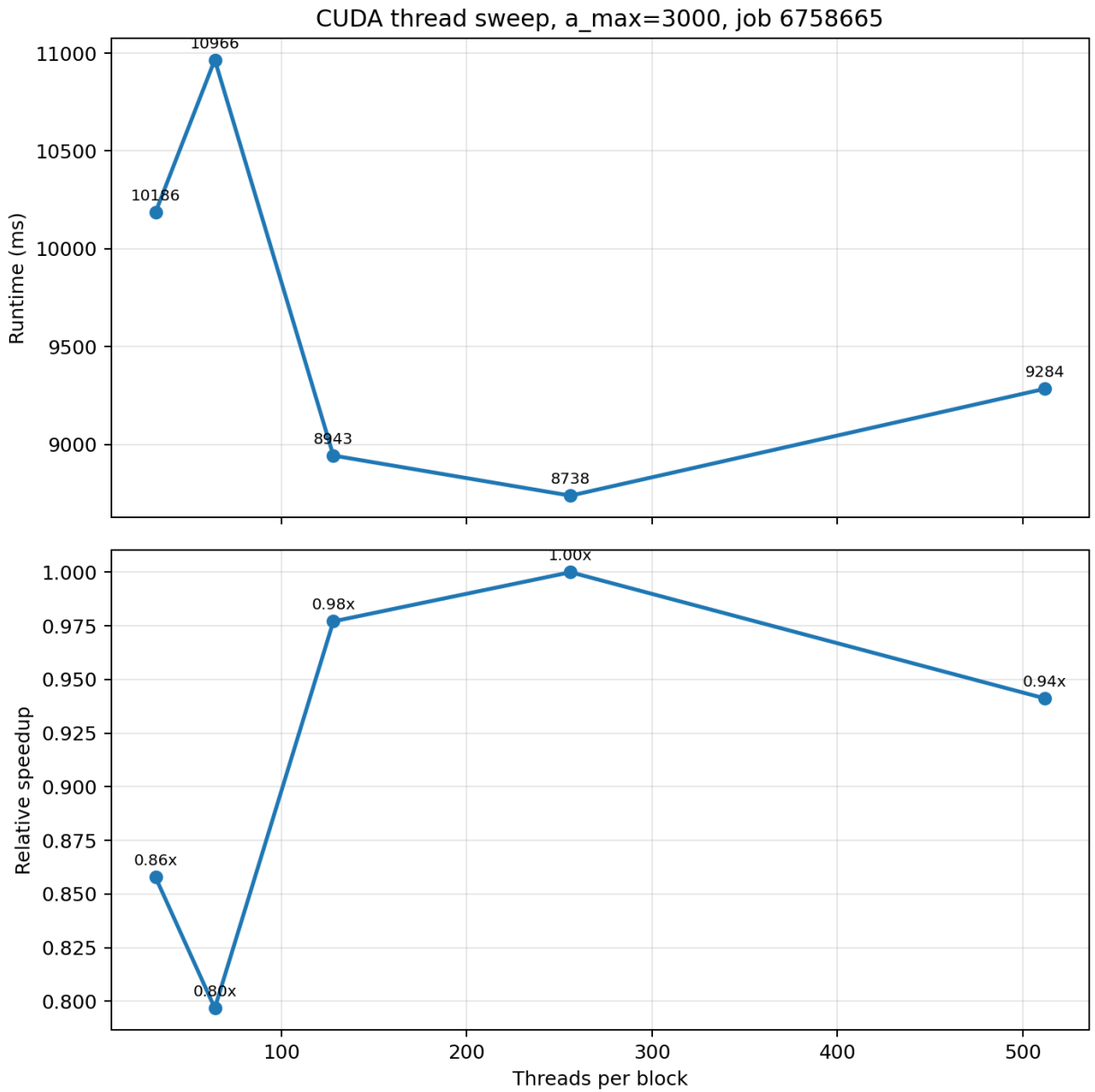
In short, CUDA parallelizes the outer (a_1, a_2) search, while modular arithmetic and precomputed tables keep each thread's inner search small.

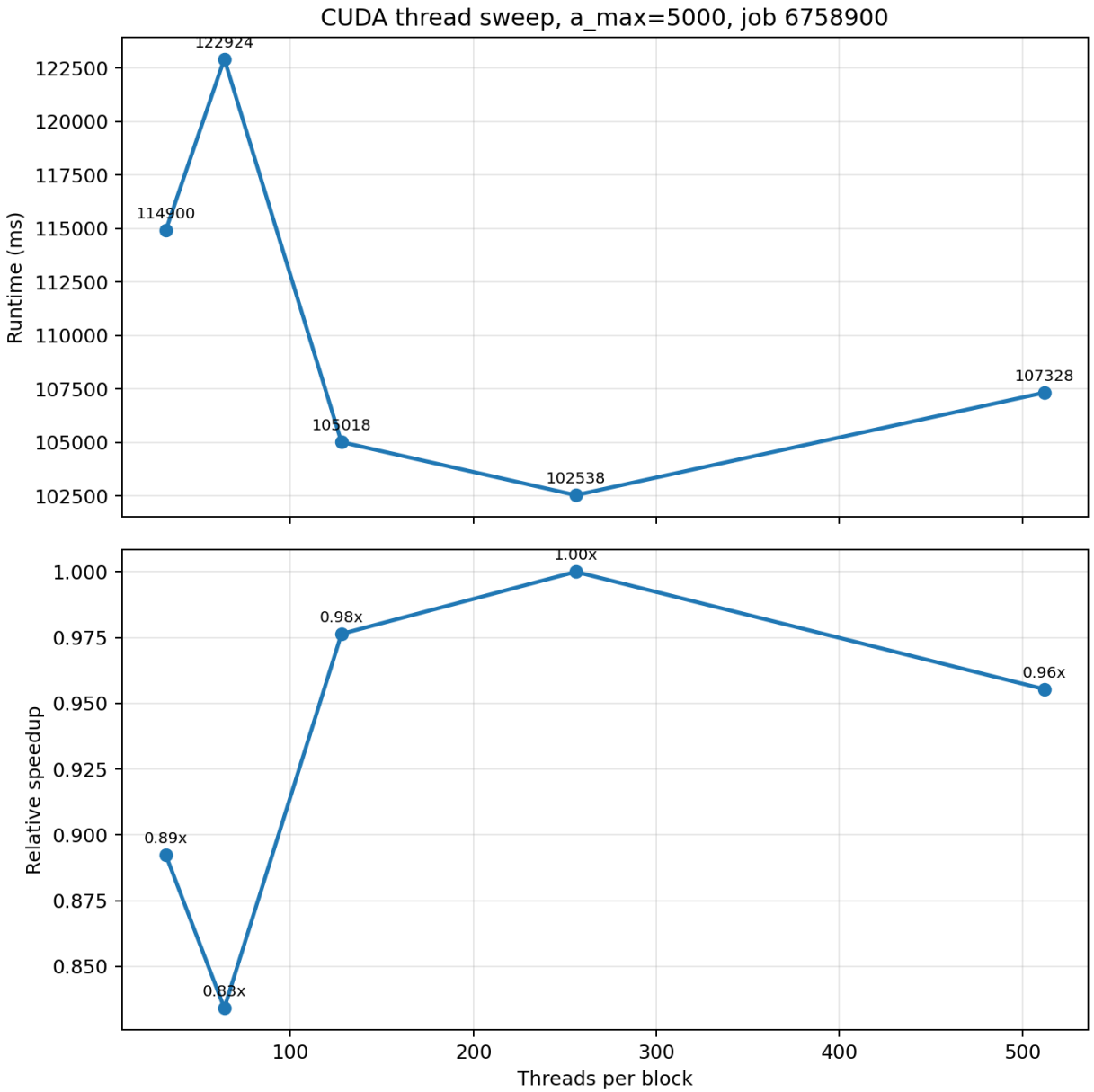
Performance: OpenMP



Performance: CUDA at $a_{\max} = 3000$



Performance: CUDA at $a_{\max} = 5000$



Performance Takeaways

- OpenMP shows strong thread scaling for this search on the CPU.
- CUDA gives lower absolute runtime, but the current kernel does not scale strongly with threads per block.
- The bottleneck is implementation quality of the GPU kernel, not the absence of parallel work in the problem itself.

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New Equation:

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Key Changes: 1 LHS term means a_1^6 is strictly congruent to 1 modulo 7. This implies that only 1 RHS term is 1 modulo 7, rest are 0. Hence,

$$a_1^6 \equiv b_1^6 \pmod{7^6}$$

avoiding $O(N)$ search for b_1.

Performance

# Threads	1	2	4	8	16	32	64	128
2 LHS, 5 RHS (ms)	136,298	71,504	36,329	18,402	9,186	4,842	2,618	1,466
1 LHS, 6 RHS (ms)	127,541	67,311	47,525	29,703	20,703	11,160	9865	10745

Figure: Execution time scaling: 2 LHS, 5 RHS ($a_{max} = 3000$) vs. 1 LHS, 6 RHS ($a_{max} = 6250$).

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Difficulty: **Scalability**

- Inner loop must resolve 5 unknown variables

No solutions found for $a_1 < 730,000$ (J.-C. Meyrignac, et al.).

References

G. Resta and J.-C. Meyrignac.
The smallest solutions to the diophantine equation $x^6 + y^6 = a^6 + b^6 + c^6 + d^6 + e^6$.
Math. Comput., 72(242):1051–1054, Apr. 2003.